

GroupGo

Smart Group-Aware Urban Transportation Decision Assistant

Mobile Application Development — Project Class 1

1. Project Name

GroupGo — Smart Group-Aware Urban Transportation Decision Assistant for Android

2. Functions

- The user can enter a starting location and a destination to receive a comparison of available transportation options (walking, public transit, ride-sharing).
- The application displays estimated travel time and cost for each transportation option.
- The application recommends the most suitable transportation option based on the user's selected preference: fastest, cheapest, or balanced.
- The user can configure a Personal Time-Value Calculator by entering their monthly salary and average working hours; the system uses this to calculate the value of the user's time per minute.
- The system incorporates the user's time value into the balanced decision algorithm to provide personalised transportation recommendations.
- The user can create a travel group and add friends to plan trips together.
- The application automatically divides the total ride-sharing cost among all group members when a group trip is recorded.
- The system stores a record of shared expenses and calculates outstanding financial balances between group members.
- The application stores trip history for each group, including route, transportation method, total cost, and date.
- Trip history and group financial balances are synchronised in real time via a cloud database, visible to all group members.
- The application automatically detects the user's current location using the device GPS sensor.
- The application displays routes visually on an interactive map, showing the selected path and estimated travel time.
- The user can log in and manage their account using Firebase Authentication.

3. Used Sensors, Hardware, and APIs

Sensor	GPS / GNSS — automatically detects the user's current device location to determine the trip origin
External APIs	Google Maps Directions API — calculates routes and travel time between locations Google Maps Geocoding API — converts addresses to geographic coordinates and vice versa Uber Ride Estimate API (or equivalent) — provides approximate ride fare and ETA
Google Services	Google Maps SDK for Android — displays interactive maps and route overlays Firebase Authentication — manages user accounts and login Firebase Firestore — stores group data, trip history, and payment records in real time
Local Database	Room Database — caches route data and stores the user's personal time-value settings for offline access

4. Complexity Level

Advanced

Justification:

- Several screens: Home / Route Search, Results Comparison, Group Management, Trip History, Expense Balances, Settings — satisfying the multiple activities / fragments requirement.
- Database required ✓ — Room Database used for local caching of route data and time-value settings.
- Internet connection required ✓ — Google Maps Directions API, Geocoding API, and Uber Ride Estimate API are used for route and fare data.
- UI: Jetpack Compose required ✓ — all screens built with Jetpack Compose.
- Architecture: ViewModel + Repository + StateFlow ✓ — full Jetpack architecture applied throughout.
- Dependency injection: Hilt required ✓ — Hilt used for injecting repositories, ViewModels, and API clients.
- Google services required ✓ — Google Maps SDK, Firebase Authentication, and Firebase Firestore are all integrated.
- Real-time cloud synchronisation ✓ — Firebase Firestore synchronises group trip history and expense balances across all members in real time.

Note: This description may be updated during the course. The complexity level and core functionality are declared at this stage; any major changes will be agreed with the instructor.